



A hillslope scale comparison of tree species influence on soil moisture dynamics and runoff processes during intense rainfall

Georg Jost^{a,b,*}, Helmut Schume^c, Herbert Hager^c, Gerhard Markart^d, Bernhard Kohl^d

^a Department of Geography, University of British Columbia, Canada

^b BC Hydro, Burnaby, Canada

^c Institute of Forest Ecology, BOKU – University of Natural Resources and Life Sciences, Vienna, Austria

^d Institute of Natural Hazards and Alpine Timberline, Austrian Federal Forest Office, Austria

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SUMMARY

This study investigates how different tree species influence soil hydrological properties that are relevant for the rainfall–runoff response of a given soil type. We hypothesize that for the same soil type, tree species that differ in rooting system, water consumption and associated soil fauna and soil flora lead to different soil moisture dynamics and lateral flow processes during rainfall and hence to different runoff responses. To test this hypothesis, we compare soil moisture patterns and interflow at different soil depths in a Norway spruce (*Picea abies* (L.) Karst) forest and in a European beech (*Fagus sylvatica* L.) forest during sprinkling experiments on two 6 × 10 m hillslope segments with the same soil type. Spruce with a shallow rooting system and sinkers that remain very shallow on poorly aerated soils and beech with a heart shaped, often deeper rooting system are two of the most important tree species in Central Europe. At each hillslope, volumetric soil water contents were measured in 6 min intervals with 48 TDR waveguides during and after sprinkling with intensities of 100 mm/h and 60 mm/h (for 1 h). The waveguides were installed in 12 soil pits, whereby a single soil pit consisted of four 20 cm buriable waveguides installed in 10 cm, 30 cm, 50 cm and 70 cm soil depth. Surface and shallow interflow at 10 cm soil depth and interflow at soil depths of 30 cm and 60 cm was automatically recorded. Despite the high rainfall intensities, no surface flow was observed in any of the experiments and only small amounts of shallow interflow were measured. Soil moisture patterns of lateral cross sections during and after the sprinkling reveal how tree species can alter runoff dynamics: under spruce, coinciding with rooting patterns, a water table develops in approximately 30 cm soil depth while the soil water content in 50 and 70 cm depth remains low. At the beech site, where coarse roots are found in deeper soil horizons, more water is directed towards deeper, already wetter soil horizons, from where the water table raises into the topsoil with high lateral conductivity. Because the higher water content on top of the stagnic layer allows segments of macropores like old root channels to connect earlier under beech, the beech hillslope exhibits a faster runoff response than the spruce hillslope. A lower water table and a higher macro-porosity makes saturation excess overland flow unlikely under beech. With the shallow water table and a lower available soil volume for preferential flow, a site planted with spruce is prone to saturation excess overland flow under natural rainfall conditions with inflow from the top. The results suggest that different tree species can lead to different rainfall–runoff responses at the same soil type. Though the study site showed minimal variation in soil properties, we cannot exclude that some of the differences in runoff processes we observed are caused by factors other than tree species, because only one large hillslope segment in each forest stand was sprinkled.

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1. Introduction

Trees influence hydrological processes that govern water flow through their canopy and their root systems. There is a vast body

* Corresponding author. Address: BC Hydro, Generation and Resource Management, Burnaby, BC, Canada V3N 4X8. Tel.: +1 604 528 1598; fax: +1 604 528 7705.
E-mail address: georg.jost@bchydro.com (G. Jost).

of literature in hydrology that compares runoff from forested watersheds to runoff from other land use types, most often clear-cuts or pastures (Alila et al., 2009; Alila and Toews, 2003; Storck et al., 1998; Thyer et al., 2004; Wigmosta et al., 2002). A common result from most of these studies is that runoff increases with decreasing forest cover (Hornbeck et al., 1997; Stednick, 1996) and that hydrographs from forested watersheds are characterized by higher base flow, lower peak flow, longer time to peak and

longer recession times. Lower antecedent soil water content because of higher evapotranspiration rates, lower net precipitation due to canopy interception, and increased macro-porosity due to tree roots and soil faunal activities (Beven and Germann, 1982; Weiler and Naef, 2003) and therefore reduced surface flow are common explanations for the different runoff characteristics of forests.

Despite the long history of studying the relationship between forests and runoff, there is little knowledge about how different tree species affect runoff, runoff processes and hydrological thresholds in a given soil type. Trees of different species modify soil properties such as soil structure and macro-porosity via their rooting system in different ways (Glatzel, 1982), which in turn can modify runoff processes and hydrological thresholds. Several studies reported of changes in soil structure and porosity due to tree roots. Radial growth of tree roots causes soil compression below coarse roots and soil uplifting on top of roots (Hartge et al., 1983). Berger and Hager (2000) found lower bulk densities in the upper mineral soil under spruce due to lifting and loosening of the soil above the root system, but higher bulk densities below the root plate. Schume et al. (2004) showed that due to higher interception evaporation, the soil under a pure spruce stand is generally drier than under a beech stand. Water flow in forested soils is not only influenced by the root systems of different tree species. Scheu et al. (2003) found significantly increased biomass of microbial and detritivorous soil animal groups under beech forests compared to spruce forests. Herbivores and predators were also more abundant under beech. From a hydrological perspective, this increased biotic activity under beech can, in addition to the effect of tree roots, result in higher macro-porosity of soils (Knight et al., 1992) and hence contribute to a different rainfall-runoff response of a beech forest compared to a spruce forests. Weiler and McDonnell (2007) described how lateral preferential flow can be explained by connectivity of pipe networks. These pipe networks created by tree roots and the soil fauna (Anderson et al., 2008; Beven and Germann, 1982; Noguchi et al., 1997) are likely to vary between species. Spatial patterns of soil water content (Jost et al., 2005) and patterns of soil properties (Zinke, 1962) are influenced by the spatial distribution of tree species. These species induced patterns could have an effect on hydrological processes that govern storm flow.

European beech (*Fagus sylvatica* L.) and Norway spruce (*Picea abies* (L.) Karst) are two of the most dominant tree species in Central Europe (Ellenberg, 1996). Spruce monocultures in Europe are extensively planted outside their natural range, often on sites where beech would be the native species (Ellenberg, 1996). Spruce is highly susceptible to windthrow and insect pest infestations, which is why many of the monocultures outside the natural range of spruce are now converted back to beech or mixed spruce-beech forests. Any conversion of forest type or simply the silvicultural selection of appropriate tree species will change the annual water balance, if evapotranspiration rates differ by directing more or less water towards streamflow. This topic is documented in the literature, though not with a vast amount of experimental work (Hibbert et al., 1967; Stednick, 1996). However, for the reasons mentioned in the prior paragraph, planting of different tree species or changes in tree species composition could also alter the rainfall-runoff response and ultimately shift the flood frequency distribution (even without affecting the mean annual flow). This is not only important in general forest management but also when dealing with climate change and forest management adaptations to climate change.

We hypothesize that tree species influence soil moisture dynamics and lateral water flow during intense rainfall. To test this hypothesis, we compare soil moisture patterns and interflow in different soil depths at a hillslope in a Norway spruce (*P. abies* (L.) Karst) forest and at a hillslope in a European beech (*F. sylvatica* L.) forest during sprinkling experiments. The soil type and the topography of the two hillslopes were the same.

2. Methods

2.1. Site and stand description

The study site is located at an elevation of 480 m above sea level on a 18% inclined north facing slope near Kreisbach, Lower Austria. In the early forties of the last century the site was planted with spruce. The current situation with a mixed transition zone of about 50 m width between a beech and a spruce stand (Fig. 1) reflects different intensities of tending and thinning during and after the end of World War II. Without human interference, natural regeneration of beech replaced planted spruce in the eastern part, while in the western part, where tending and selective thinning were fully practiced from the beginning, a pure spruce stand developed. The natural plant association at this site is an *Asperulo odoratae* – *Fagetum*. Canopy closure in both stands is around 90% and LAI is so high (7.1 in the spruce stand and 7.9 in the beech stand) that no shrub and hardly any forest floor vegetation exist. The mean annual precipitation of 850 mm is usually sufficient for both species unless a severe summer drought occurs (like it occurred in the years 2000 and 2003). The mean annual temperature is 8.4 °C.

The soil type at Kreisbach is a Stagnic Cambisol, which developed from Flysch sediments (sandstones, marls, mudstones, and claystones). The soils exhibit a high clay content. In such cohesive soils, tree roots are an important factor in soil structure formation and therefore influence hydrologically important soil properties such as macro-porosity and bulk density. For example, as it will be shown later in the Results, bulk density increases as root density decreases with depth. At about 60–70 cm depth, the water infiltration rate becomes very low, periodically causing water logging and anaerobic conditions. This causes hydromorphic features in all soil horizons with Fe/Mn-concretions in the top 50 cm and mottles in deeper layers. To study the tree species effect with minimal variation between soils, several sites with adjacent beech and spruce stands were screened with 15 randomly selected soil samples in each stand. With the objective to isolate the tree species effect on runoff from other controls on runoff processes, the Kreisbach site was chosen, because it showed very little variation in morphological soil properties, like thickness of mineral soil layers, texture class or soil structure between and within the stands.

Due to species specific aeration requirements, beech and spruce develop different rooting patterns in the Stagnic Cambisols (Schmid and Kazda, 2001). At the same site, Schmid and Kazda (2001) analyzed rectified photographs of 40 m² of soil profile walls for coarse root density in both spruce and beech stands using a computer aided mapping technique. They found that spruce root density shows an exponential decrease over depth, while beech root density follows a gamma function with a maximum between 10 and 20 cm soil depth. In both monospecific stands, roots were found down to 100 cm depth. By splitting the roots into 3 size classes, they found that there are no large spruce roots (>20 mm) below 20 cm depth and that the density of medium sized coarse roots (6–20 mm) is on average 31% higher under beech below 10 cm depth.

2.2. Soil physical parameters

After the sprinkling experiments (see Section 2.3), a total of 24 soil cores (200 cm³; 7.2 cm in diameter and 4.9 cm long) were taken from three soil pits at 10, 30, 50, and 70 cm depth in each stand (two cores at each depth, one in vertical direction one in horizontal direction). The sampling depths corresponded to the depths of water content measurements described in the next section. Samples that contained stones bigger than 40 mm or roots bigger than 20 mm were discarded in the field. This sampling underestimates the total root mass and the coarse soil volume as well as larger

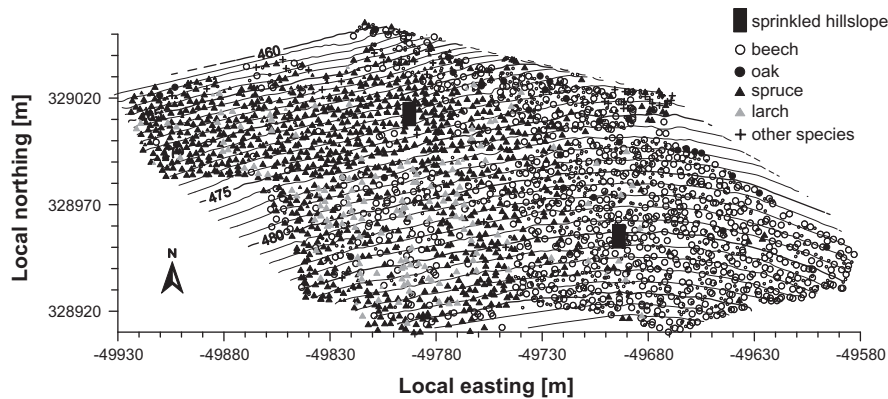


Fig. 1. Tree species distribution and location of the sprinkled hillslope segments at the study site.

macropores. The saturated hydraulic conductivity was measured in the laboratory by applying a permanent pressure head of 10 mm water column to the saturated samples for at least 1 h. The outflow was recorded continuously. According to Austrian standards (Ö-Norm L-1065), the first 15 min of the measurements were discarded. The saturated hydraulic conductivity was calculated as the arithmetic mean of the subsequent five stable values. After measuring the saturated hydraulic conductivity, the samples were sieved (mesh size 2 mm) to obtain the root, coarse and fine soil fractions. The oven dry root mass was obtained by drying the root fraction at 105 °C. Bulk density of the fine soil and pore volume were calculated from the oven dry fine soil fraction.

The texture class of the fine soil samples (<2 mm) was analyzed from one homogenized mixed sample per stand and soil horizon by wet sieving down to a particle size of 40 µm. Below that fraction a particle size analyzer (Shimadzu SA-CP2) was used.

2.3. Experimental setup

To test the influence of tree species on soil moisture patterns and lateral flow processes, a 6 × 10 m hillslope segment was sprinkled with intensities of approximately 100 mm/h and 60 mm/h for a duration of one hour at both the beech and the spruce stand. At each hillslope surface and shallow interflow at 10 cm soil depth and interflow at soil depths of 30 cm and 60 cm were collected from a trench by inserted metal sheets and gutters and automatically recorded. The maximum depth of 60 cm for interflow measurements was chosen, because we expected negligible interflow within the dense stagnic soil layer below. Sprinkling experiments were carried out on 26th September 2001 in the spruce stand and on 27th September 2001 in the beech stand. Both hillslopes were first sprinkled with 100 mm/h and then, when outflow had stopped (after ~2 h), with 60 mm/h. Each experiment lasted for 1 h except for the 60 mm/h experiment on the spruce hillslope, which lasted only 56 min. After starting the 60 mm/h sprinkling experiment in the beech stand, the metal sheets that collected outflow in 30 and 60 cm soil depth were washed out. As a consequence, we collected interflow in this particular experiment from the bottom of the soil pit at 90 cm depth, while still measuring surface flow and shallow interflow at 10 cm soil depth. Time series of runoff coefficients K for each hillslope were calculated as

$$K = \frac{Q}{P} \quad (1)$$

where P is the sprinkling intensity in mm/h and Q is the runoff intensity collected in the trench prorated to mm/h for each measurement interval (the outflow was measured in irregular intervals depending on the time to fill the collection buckets to a certain volume).

In order to obtain high resolution time series of soil water content in the hillslope segments during and after sprinkling, we equipped 12 soil pits at each site with TDR wave-guides that were automatically logged (setup see Fig. 2). In each soil pit 20 cm buriable TDR wave-guides were installed in four soil layers ranging from 0–20 cm, 20–40 cm, 40–60 cm and 60–80 cm soil depth. Wave-guides were inserted in 45° angles centered around the mid-points of these soil layers (dashed lines in Fig. 2). For simplification, we refer to these measurements with the midpoints of the respective layers in the following text (i.e. 10, 30, 50, and 70 cm). To allow settling of the soil around the TDR wave-guides, the wave-guides were installed 3 months prior to the experiments. The spacing between the soil moisture measurement pits was roughly 2 m along the slope and 1.5 m perpendicular to the slope direction. Three Trase TDR-systems (Soilmoisture Equipment Corp.), each one multiplexing 16 wave-guides, were used to measure the volumetric water content (VWC) in 6 min intervals. In order to directly compare soil water dynamics between the spruce and the beech site by means of mass balance calculations, VWC was converted to soil water storage (SWS) in mm, whereby the volume of the coarse soil fraction (>2 mm) in each layer was subtracted from the total soil volume. To address some of the uncertainties associated with the hillslope experiments, mass balances were calculated from the measured VWC and from runoff measurements. For every time interval i we can calculate ΔSWS_i relative to the initial conditions from our known inflow I and measured outflow O by using the mass balance

$$\Delta SWS_i = \sum_{t=1}^i I_t - \sum_{t=1}^i O_t \quad (2)$$

Measuring at the same time SWS by means of TDR allowed us to derive such relative changes in SWS also from these measurements (ΔSWS_{meas}). The difference between measured and calculated ΔSWS is an error that, if negative, points to a water loss that has not been captured in the collection trench, and if positive, indicates that there was water input that has not been captured by the TDRs. The overall mass balance error also comprises measurement errors.

3. Results

3.1. Soil physical measurements

A two sided t -test revealed that horizontal and vertical soil samples in both hillslopes showed no significant differences in soil physical parameters. Therefore, both sets of samples were pooled resulting in a sample size of six for each soil horizon in each hillslope.

While roots of both beech and spruce are found down to 70 cm soil depth, the roots of spruce are more abundant in the uppermost

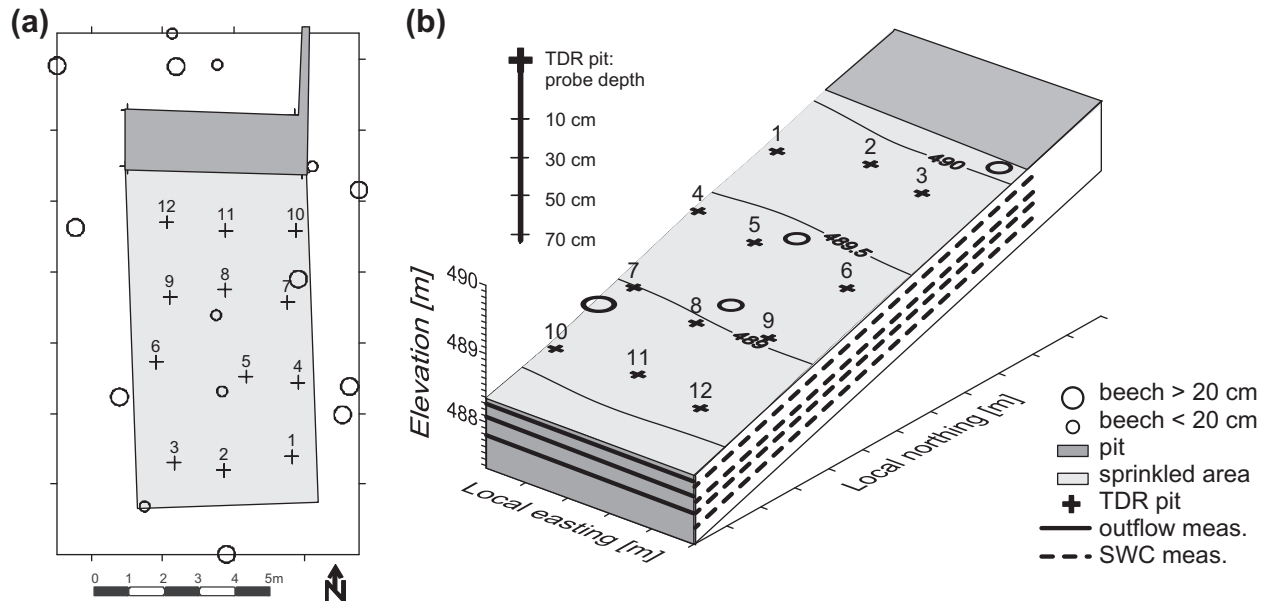


Fig. 2. Plan view (a) and 3-dimensional view (b) of the sampling design in the sprinkling experiments.

soil horizon (Fig. 3a). Although Fig. 3a suggests that beech and spruce have a different root distribution over depth, with spruce having a higher root concentration in 10 cm and beech in 30 cm depth, those differences are not statistically significant (for the deepest horizon the t-test result was close to significant). However, the observed distributions over depth are in line with those described by Schmid and Kazda (2001). As stated in Section 2, our sampling method underestimates the hydrologically important coarse roots (>20 mm).

The fine soil bulk density in both hillslopes increases over depth from 970 kg/m^3 in 10 cm to 1390 kg/m^3 in 70 cm depth (Fig. 3b) with corresponding pore volumes ranging from 59% in the topsoil to 43% in the subsoil (Fig. 3c). The high abundance of spruce roots in the uppermost horizon resulted in a significantly lower bulk density of the fine soil under spruce compared to beech (Fig. 3b) and therefore also in a significantly higher pore volume (Fig. 3c). The rock content (not shown) varies between 3 and 9 vol.%. Soils in both stands are heavily textured with a clay content that slightly increases from 20% in the topsoil to 30% in the subsoil. Seven out of eight samples are categorized as silt loam according to the USDA classification, one sample is categorized as a silt clay loam.

The mean saturated hydraulic conductivities of the soil layers ($n = 6$ in each layer) range between 0.8 and 27 m per day. Under spruce the highest conductivity can be found in 10 cm soil depth, while under beech conductivity is highest in 50 cm soil depth. At both sites the deepest soil horizons showed the lowest values with 0.8 and 2 m/d for beech and spruce, respectively (Fig. 3d). Due to the high variability, none of the differences is statistically significant.

3.2. Soil water dynamics during and after sprinkling

Results in this section focus on the sprinkling experiments with 100 mm/h intensities. The dynamics in the 60 mm/h experiments were basically the same as in the 100 mm/h experiments.

Due to a rainy September the antecedent VWC in both hillslopes was relatively high compared to the average VWC in the growing season. Prior to the 100 mm/h (60 mm/h) experiments, VWC was 2.1 vol.% (1.2 vol.%) lower under spruce compared to the beech site. The mean VWC over all TDR's prior to the 100 mm/h (60 mm/h) sprinkling was 33.9 vol.% (38.8 vol.%) under spruce and 36.0 vol.%

(40.0 vol.%) under beech (Table 1). The spruce and the beech hillslope segments exhibited different soil water dynamics during and after sprinkling: in the spruce hillslope most of the soil water changes occurred in the topsoil (10 and 30 cm soil depth), while the water content in the subsoil (50 and 70 cm soil depth) showed only little variation during and after sprinkling (Fig. 4a). Mean VWC under spruce increased by 9.9 vol.% in the 100 mm/h experiment and by 4.1 vol.% in the 60 mm/h experiment. Due to the higher antecedent VWC, the mean VWC under beech increased by only 6.5 and 1.8 vol.% for the 100 and 60 mm/h experiments, respectively (Table 1). Despite the overall lower increases in mean VWC in the beech hillslope, VWC increases in 50 cm soil depth were higher under beech (Table 1 and Fig. 4b). Only small VWC increases in the stagnic soil horizon at 70 cm soil depth were found in both stands (Fig. 4). In both stands the deeper (30–70 cm) soil layers responded already 12 min after the start of sprinkling, which indicates infiltration through vertical macropores in both hillslopes.

It is difficult to analyze soil water dynamics from VWC alone, because porosity in both hillslopes decreases with depth (Fig. 3c). Changing porosity with depth can be accounted for by calculating the available porespace in each soil layer (difference between porosity from soil physical measurements and VWC measured by the TDR's). Dynamics of available pore space during and after sprinkling reveal distinct patterns between the two hillslopes (Fig. 5); under spruce a saturated layer developed at about 30 cm soil depth, from where a perched water table rose towards the surface. At the end of the 100 mm/h sprinkling experiment, the whole topsoil under spruce was close to saturation (Fig. 5a). Water in the beech hillslope moved vertically all the way down to the stagnic layer, and the soil profile got saturated from the bottommost soil layer. Even after one hour of sprinkling with 100 mm/h, all soil layers except the stagnic layer at 70 cm soil depth were far below saturation. Towards the end of the sprinkling, likely a result of flow accumulation or the trench itself (see Section 4), less available pore space (or higher VWC in Fig. 4) was found near the trench than in the uphill sections of both hillslopes (Fig. 5). Under spruce this effect appeared mainly in the top two soil layers, whereas under beech it was visible only in the deepest soil layers at the bottom of the investigated hillslope segment.

The sprinkling experiments with 60 mm/h (subsequent to the 100 mm/h intensities) showed basically the same patterns of

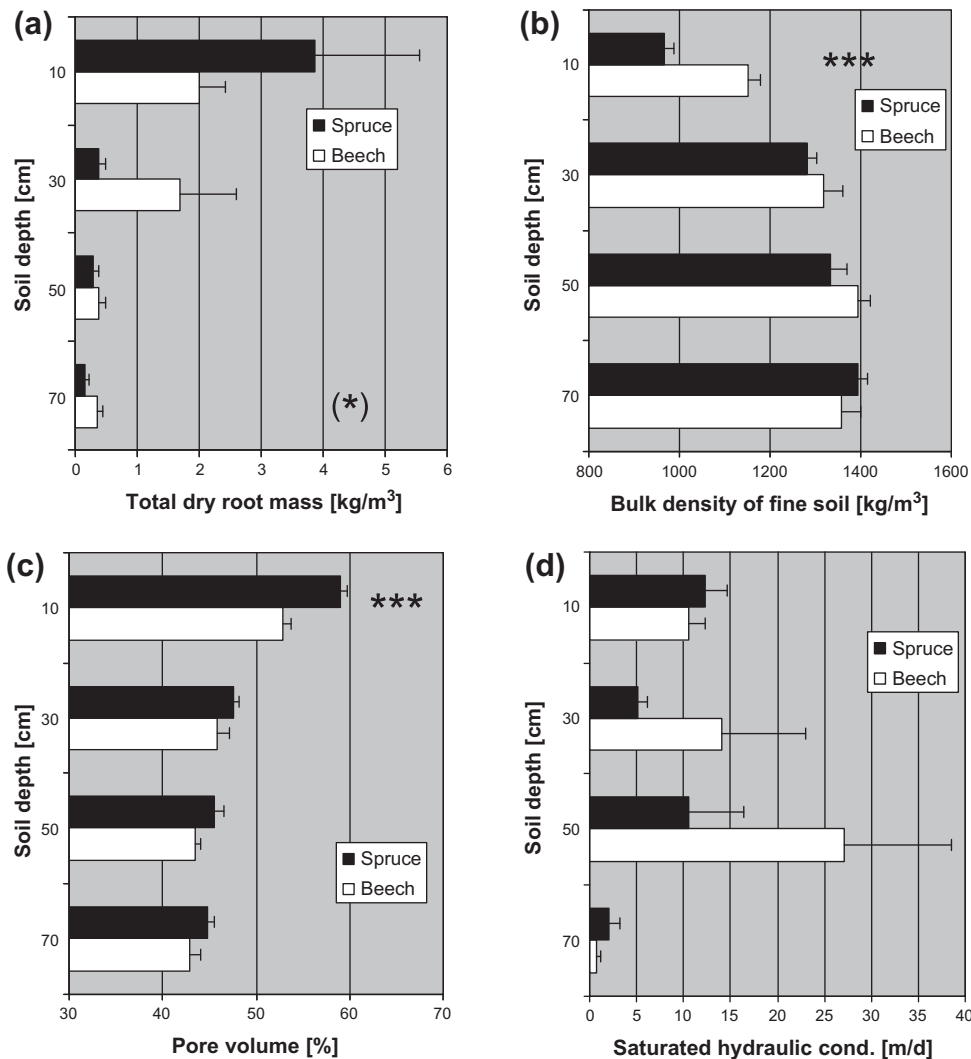


Fig. 3. (a) Total dry root mass, (b) bulk density of the fine soil, (c) pore volume and (d) saturated hydraulic conductivity over depth by hillslope. The error bars represent the standard error of the mean. The stars indicate the following significance levels in the applied *t*-tests: *** $P < 0.001$, ** $0.001 < P < 0.01$, * $0.01 < P < 0.05$, (*) $0.05 < P < 0.1$.

VWC and changes in remaining air filled pore space with the exception that the topmost soil layer in the spruce hillslope was not as close to saturation as in the 100 mm/h experiment. The antecedent VWC prior to the 60 mm/h experiments was close to field capacity in both hillslopes (Table 1).

3.3. Runoff and interflow measurements

At both hillslopes all applied water infiltrated and neither infiltration excess overland flow nor saturation excess overland flow occurred during any of the 100 mm/h and 60 mm/h sprinkling experiments. The total runoff and interflow in all soil layers were substantially higher in the beech hillslope (Table 1). As it will be shown later with mass balance calculations, the quantification of observed runoff has to be interpreted with caution.

The hydrographs from the spruce and the beech hillslope (Fig. 6), given as runoff coefficients (Eq. (1)), show some distinct features that can be seen in both the 100 mm/h and the 60 mm/h experiments (The peaks in the 60 mm/h experiments are slightly offset due to differences in sprinkling duration – see Table 1). The hydrographs of total runoff at the beech hillslope exhibit steeper rising limbs and faster response times compared, to the spruce hillslope. In both experiments, the beech hillslope responded about

6 min faster than the spruce hillslope. For the 100 mm/h experiment (Fig. 6a), beech also showed a steeper recession. Recessions of the 60 mm/h experiments (Fig. 6b) cannot be directly compared, since runoff in the beech hillslope was collected in 90 cm soil depth and because, as will be shown later, the mass balances between the two hillslopes are very different. Towards the end of both sprinkling experiments, though less pronounced in the 100 mm/h experiment, the beech hillslope reached a constant flow rate, while runoff in the spruce hillslope rose steadily with sprinkling duration.

3.4. Cumulative mass balances

Under the assumption of no flow boundary conditions through the dense, water logging soil layer at 60 cm depth as well as through the top and through all sides of the hillslope except for the trench wall, the ΔSWS calculated according to Eq. (2) from the cumulative volume of the applied water (gray lines in Fig. 7) minus the cumulative measured outflow (solid gray areas in Fig. 7) should equal the measured ΔSWS at any point in time. Any negative diversion between measured ΔSWS (solid black lines in Fig. 7) and calculated ΔSWS (dashed black lines in Fig. 7) points to unaccounted water losses (gray dashed lines in Fig. 7). All four

Table 1

Summary statistics of sprinkling experiments at the spruce and beech hillslope. Volumetric water content (VWC) statistics along the soil profile are based on averages of 12 TDR sensors in each soil depth.

	Spruce		Beech	
Sprinkling intensity (mm/h)	115	61	109	66
Duration (h:mm)	1:00	0:56	1:00	1:00
<i>Antecedent VWC (vol.%)</i>				
10 cm	31.0	36.3	31.4	37.2
30 cm	32.5	41.6	33.7	38.7
50 cm	34.2	37.8	36.5	40.6
70 cm	38.0	39.5	42.3	43.4
Mean	33.9	38.8	36.0	40.0
<i>Peak VWC during sprinkling (vol.%)</i>				
10 cm	52.0	48.7	45.4	42.7
30 cm	45.8	45.3	40.5	40.1
50 cm	37.8	37.9	40.7	40.8
70 cm	39.5	39.6	43.4	43.6
Mean	43.8	42.9	42.5	41.8
Δ VWC, peak – start (vol.%)				
10 cm	21.0	12.4	14.0	5.4
30 cm	13.3	3.7	6.8	1.5
50 cm	3.6	0.1	4.2	0.2
70 cm	1.6	0.1	1.0	0.2
Mean	9.9	4.1	6.5	1.8
<i>Runoff (% of input)</i>				
0–10 cm	6.3%	7.6%	15.3%	12.8%
10–30 cm	7.2%	8.2%	13.6%	Broken
30–60 cm	10.1%	12.9%	11.1%	Broken
10–90 cm				74.5%
Total (%)	23.4	28.5	40.0	87.3
<i>VWC statistics – mean over all soil layers (vol.%)</i>				
Highest VWC during sprinkling	43.8	42.9	42.5	41.8
VWC 1 h after end of sprinkling	40.1	40.1	40.2	40.2
Δ VWC peak – start	9.9	4.1	6.5	1.8
Δ VWC 1 h after end – start	6.2	1.3	4.2	0.2
<i>Water loss</i>				
End of sprinkling (%)	–10	–21	–11	–14
Recession trench (%)	–41	–69	–26	–4
Recession TDR (%)	–17	–21	–16	–15

experiments with the exception of the 60 mm/h experiment under beech show water losses that gradually increase over the course of sprinkling and during the recession (Fig. 7), which means that some of the inflow was neither found in the outflow measurements nor as increase in SWS and hence a no flux boundary condition was violated.

In all four experiments the water losses were more or less zero up to the point, where the first runoff was measured in the trench. This shows that more or less the entire water input was captured by the 48 TDR probes in each hillslope segment. A lag in TDR response plus the fact that all 16 waveguides from each of the three TDR pulsers were given the same time stamp cause the mass balance to be slightly off at the beginning in all experiments. From the beginning of the outflow until the end of sprinkling, the water losses in all experiments are still in an acceptable range of –21% relative to input rates or less negative (–10% and –21% for the spruce hillslope and –11% and –14% for the beech hillslope for 100 mm/h and 60 mm/h, respectively; see Table 1). However, during the runoff recession mass balances in all experiments, except in the 60 mm/h experiment at the beech hillslope, became increasingly negative due to water losses (Fig. 7, Table 1). The growing gaps between the hydrograph recessions and the decreases in SWS_{meas} after the end of sprinkling show that in three of our experiments only a small part of the TDR measured soil water storage losses were collected in the trenches. The 60 mm/h experiment at the beech hillslope (Fig. 7d), the one where interflow was collected down to 90 cm soil depth (because metal sheets were washed out, see above), is an exception and offers some explanation

as to where water was lost during the experiments. In this experiment the mass balance was more or less closed throughout the entire experiment including the runoff recession. The most obvious explanation for water losses is that there was substantial interflow below 60 cm soil depth within the stagnic layer that was not captured with the measurements down to only 60 cm soil depth in the other experiments. This might be true for the beech hillslope. In the spruce hillslope, the VWCs of the two soil layers at the bottom (50 and 70 cm) were still far from saturation in both experiments, so that it is unlikely that substantial (saturated) water flow in the soil matrix occurred within these horizons. Without saturated soil matrix, substantial amounts of preferential flow are also unlikely, since the hydraulic gradient would point towards the soil matrix. Also, massive amounts of interflow in the stagnic layer were not visible in either of the other three experiments. Hence, in the spruce hillslope it is more likely that the negative mass balances were caused by water losses through the sides of the hillslope.

The shape of the mean SWS_{meas} time series offers (Fig. 7) some explanations, why hydrographs under spruce and beech look so different. Under spruce, the mean SWS rose until the end of both sprinkling experiments. This shows that water was still going into storage until the end of sprinkling. The consequence is a hydrograph with a steadily increasing rising limb (Fig. 6). Under beech, the SWS reached a sill after about 30 min in the first (100 mm/h) experiment and after about 15 min in the second experiment, which indicates steady state flow conditions. It is noteworthy that the VWCs during the steady state flow conditions under beech were substantially below the saturated VWCs in all but the bottommost soil layer (see also Fig. 4b).

Another difference between the two hillslopes is the rate by which the SWS decreased after the end of sprinkling and the shape of the SWS recession. In the spruce hillslope the SWS decreased much slower and more steadily than in the beech hillslope. Since the mass balance error was low in all experiments until the end of sprinkling, the differences in recessions deduced from TDR measurements of SWS might be more representative of the tree species influences on hydrology than the recession of the runoff measured in the trench. SWS in the beech hillslope initially decreased very fast and slowed down after about 30 min. By this time the SWS in the 60 mm/h experiment under beech was almost at its initial value (field capacity). SWS recession under spruce did not exhibit this sharp bend. Under spruce, SWS decreases more steadily. In the 60 mm/h experiment under spruce, SWS reached its starting value, which was around field capacity, only after about 2 h.

3.5. Approximated runoff with recession from changes in VWC

As pointed out in the prior section, the recessions calculated from measured SWS might be more representative of the characteristic tree species influences on soil hydrology than the runoff measured recessions. To allow a more direct comparison of tree species effects on the hydrographs, we replaced the runoff measured at the trench wall by the runoff calculated from the decrease in SWS from the first time step after the sprinklers had stopped onwards. We are aware that this is only an approximation and results should be interpreted in a rather qualitative than quantitative way. Recessions calculated from TDR measurements of SWS are independent from the flow direction of water (sideways, downslope or percolation) and thus the mass balances are less negative as compared to the recessions calculated from the runoff measured at the trench wall. After replacing the measured runoff recessions with SWS decreases per time step, the mass balance errors are –17% and –21% for the 100 mm/h and 60 mm/h experiments at the spruce hillslope and –16% and –15% for the 100 mm/h and 60 mm/h experiments at the beech hillslope (Table 1).

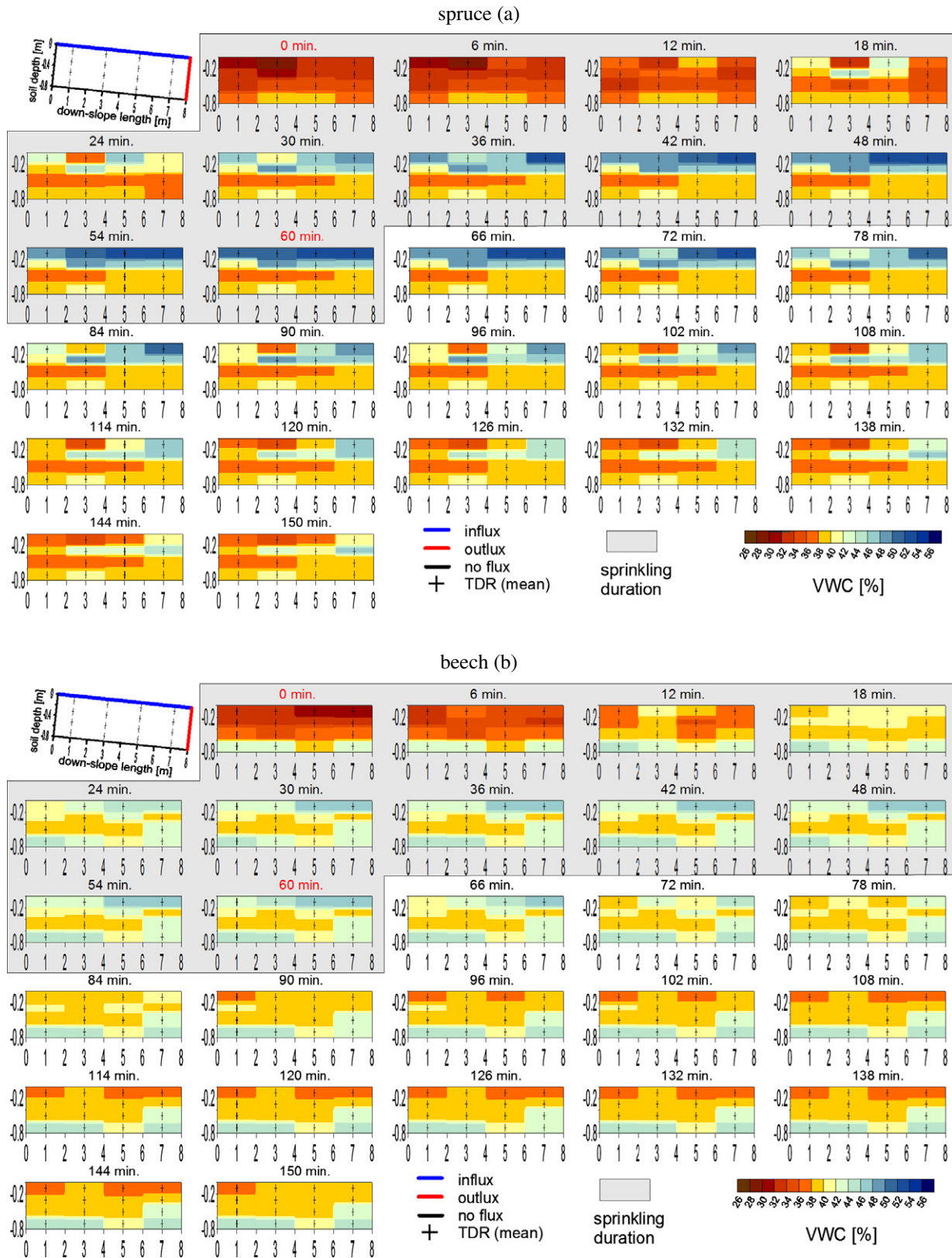


Fig. 4. Time series of spatial patterns (depth over slope length) of soil moisture during and after sprinkling with 100 mm/h in the spruce (a) and beech (b) hillslope derived from nearest neighbor interpolations with 6 min time difference. Each cross marks the mean of three TDR wave-guides at the same depth and the same slope length. Note that slope length is measured from the top of the hillslope to the lowest TDR, which is 2 m upslope from the collection trench.

Fig. 8 shows hydrographs for all four experiments with the original recessions from runoff measurements as solid lines and

hydrographs with SWS approximated recessions in dotted lines. Note that because of the coarser time interval of SWS sampling

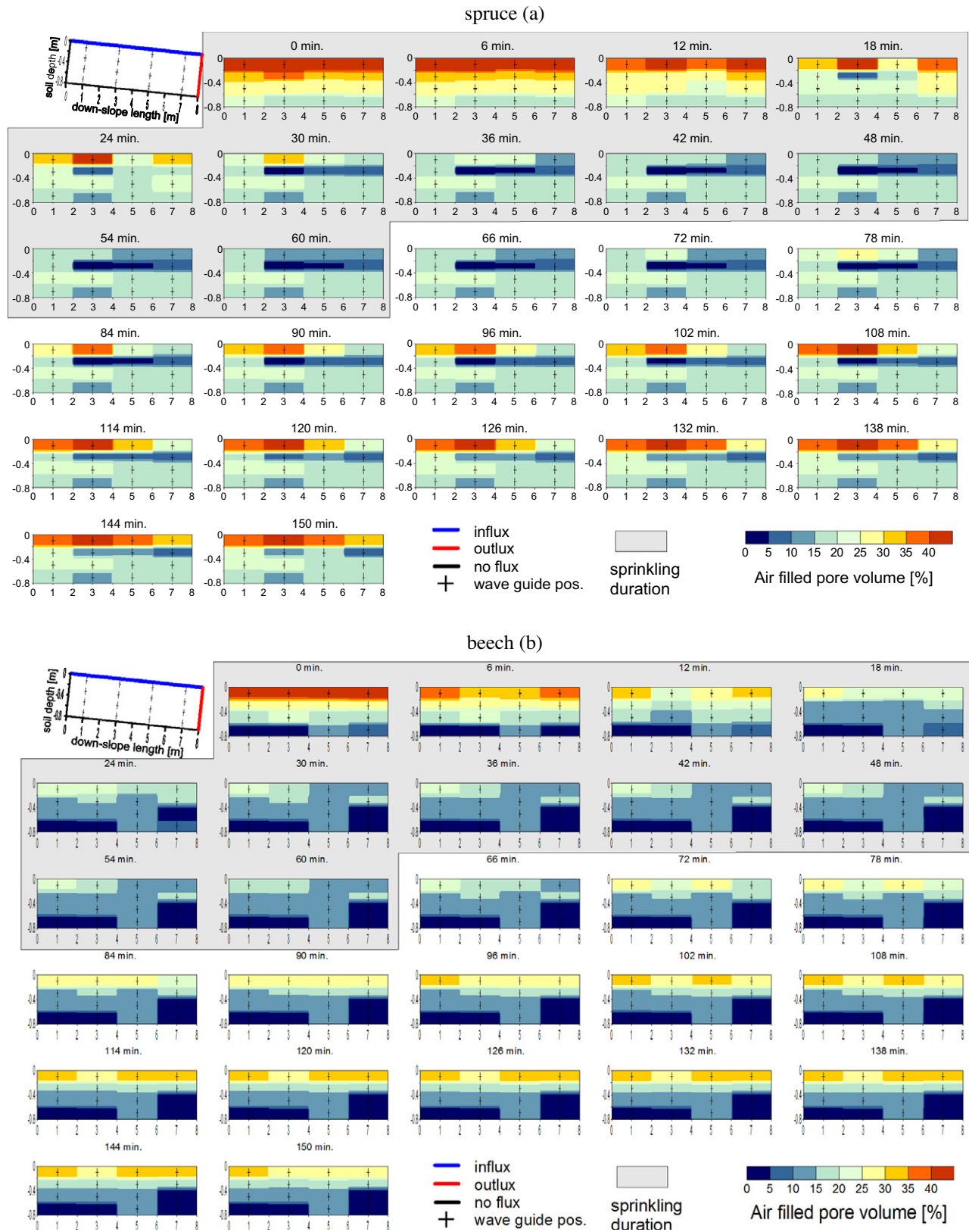


Fig. 5. Time series of spatial patterns (depth over slope length) of remaining air filled pore space during and after sprinkling with 100 mm/h in the spruce (a) and beech (b) hillslope derived from nearest neighbor interpolations with 6 min time difference. Each cross marks the mean of three TDR wave-guides at the same depth and the same slope length. Note that slope length is measured from the top of the hillslope to the lowest TDR, which is 2 m upslope from the collection trench.

(6 min), the rising limbs in Fig. 8 appear different to those given in Fig. 6. The hydrographs in Fig. 8 summarize what was found in the

previous sections: under beech, the time for runoff initiation is shorter, the rising limb is steeper, runoff exhibits a higher peak

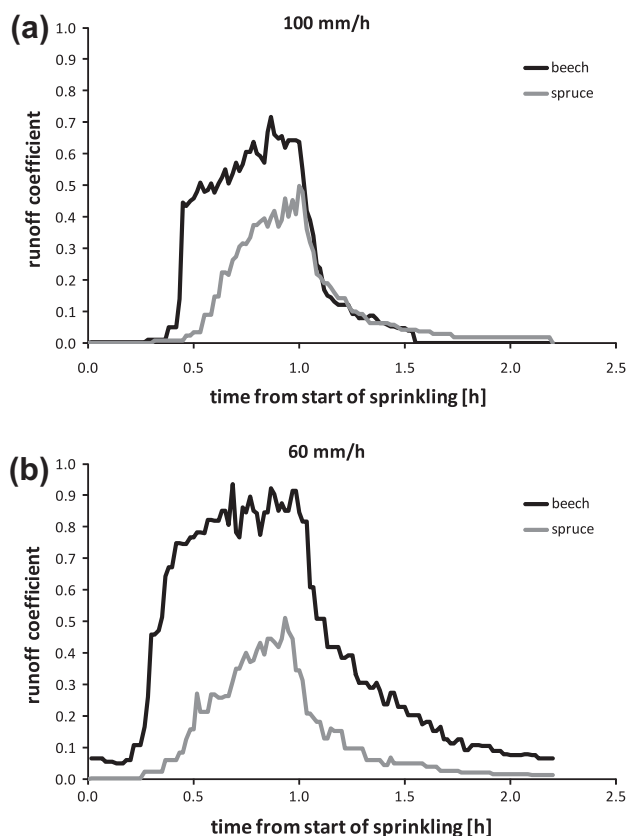


Fig. 6. Measured runoff coefficient for total runoff at hillslope segments in a spruce and a beech forest during and after sprinkling with intensities of around 100 mm/h (a) and 60 mm/h (b). The exact intensities are given in Table 1. Note that in the 60 mm/h experiment at the beech hillslope the collection sheets were washed out and interflow was therefore collected from the bottom of the collection trench at 90 cm soil depth.

and recesses faster. Under spruce, the time for runoff initiation is longer, the rising limb is less steep, runoff shows a smaller peak and recesses substantially slower.

4. Discussion

The findings of runoff and interflow measurements, particularly the recession limbs obtained from changes in SWS and those seen in the hydrographs from direct runoff measurements, do not give a congruent picture of the exact influence of spruce and beech on runoff processes. Hence, both TDR and runoff measurements should be seen as two parts of the same story that need to be put together by using the mass balances. In the following, we first discuss soil water dynamics and runoff measurements separately and then attempt to create a full picture of the influence of spruce and beech on runoff processes.

4.1. Soil water dynamics during and after sprinkling

Lower antecedent soil water content such as found before the 100 mm/h experiments is typical for spruce. At the same site, Schume et al. (2004) found that due to higher canopy interception, VWC under spruce is generally lower and shows smaller fluctuations throughout the growing season. This increases the water absorption capacity in the topsoil of the spruce stand compared to the beech stand and offers one explanation for the longer runoff initiation time under spruce. Dry, water repellent conifer litter can be excluded as a retarding factor for water infiltration in Kreisbach, since the humus

form in the spruce stand was an only 2 cm thick mull humus. Under spruce a perched water table develops closer to the soil surface than under beech (Fig. 4). The depth of 30 cm, at which the perched water table develops under spruce, coincides with low measured saturated hydraulic conductivities (Fig. 3d). At the beech hillslope, percolation is markedly restricted only at 70 cm depth, while the mean saturated hydraulic conductivities of the 30 and the 50 cm layers were the highest of all (Fig. 3d). Under beech, the bigger soil volume available for water absorption compensates the enhanced water absorption capacity of the topsoil under spruce.

Since the response times of the deep TDR sensors in both hillslopes are still by a factor of ~ 10 faster than the highest measured saturated hydraulic conductivities, infiltration must be dominated by preferential flow such as flow in macropores of old root channels and in pores created by rodents, earthworms and other animals (Beven and Germann, 1982; Weiler, 2005; Weiler and Naef, 2003). Because the antecedent water content in all experiments was below or at field capacity, the vertical preferential flow is likely initiated at the soil surface.

Despite this similarity, the spruce and the beech hillslope differ in their vertical water recharge patterns. In accordance to the vertical distribution of medium sized and large coarse roots (Schmid and Kazda, 2001), the topsoil under spruce shows high recharge rates that bring the water content close to saturation, while the subsoil remains relatively dry compared to the beech hillslope. Coarse roots fragment the soil, and thereby create soil structure and drainage pathways. After their decay, roots leave pathways for preferential flow. Infiltration through vertical macropores from old beech roots (Beven and Germann, 1982) causes water in the beech hillslope to reach the stagnic layer within minutes. A water table then develops on top of the stagnic layer. These infiltration differences between the tree species – beech more to the bottom and spruce more into the topsoil – is not so pronounced under very dry soil conditions. Jost et al. (2004) showed that after soil drought soil water recharge started at a similar depth of approximately 60 cm under both species, due to the formation of vertical shrinkage cracks in the clayey soils at Kreisbach. Dekker and Ritsema (2000) described a similar phenomenon. However, once shrinkage cracks are closed, the differences in lateral flow between the spruce and the beech site are likely to be similar to those observed in this study.

Similar to vertical flow velocities, the lateral flow velocities are higher than those calculated from the saturated hydraulic conductivities of the soil matrix. Hence, preferential flow in macropores has to be the dominant flow process in both forests also in the lateral direction. Water flow in macropores can be initiated in two basic ways (Weiler and McDonnell, 2007): from the soil surface or from subsurface. For subsurface initiation of macroporous flow, i.e. for water to move from the soil into the macropore, the matrix potential of the soil must be at or close to saturation. Several observations favor the idea that the lateral macropore flow in both hillslopes is predominantly initiated at the subsurface and not at the soil surface. If lateral flow in macropores was initiated at the soil surface, the hydraulic gradient would point from the macropore towards the soil matrix, since the matrix potential of the dry soil is more negative than in the macropore, where water is only under atmospheric pressure. As a result, the macropore would become drier in downslope direction and only little water would reach the trench. Another sign that favors subsurface initiation of macropore flow is the observation that runoff starts when one of the soil layer gets saturated. In the beech hillslope this is the first layer on top of the stagnic layer at 50 cm depth, while in the spruce hillslope it is the layer right underneath the dense network of coarse roots (Schmid and Kazda, 2001) at 30 cm depth.

Subsurface initiation of macropore flow helps to explain the different rising limbs of the hydrographs and the different runoff initiation times under spruce and beech. In the beech hillslope VWC

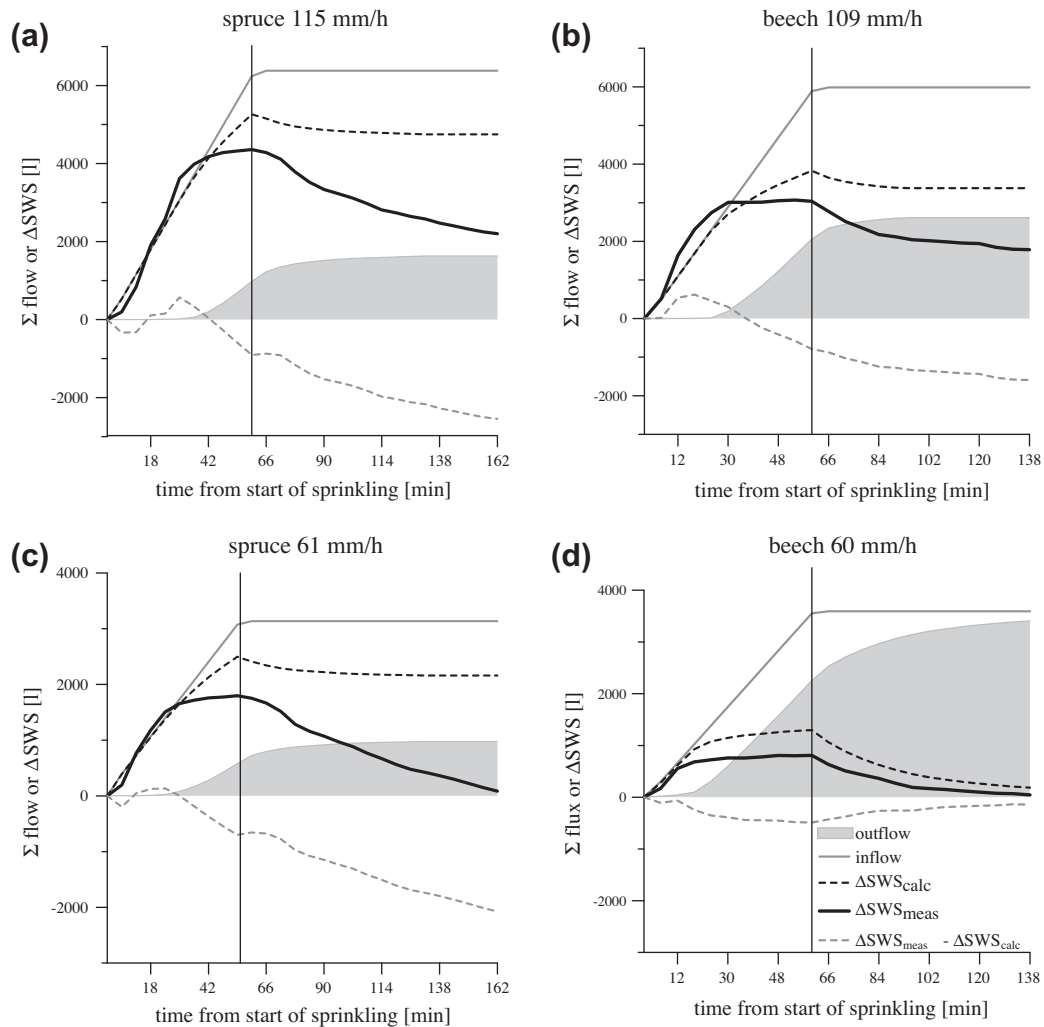


Fig. 7. Cumulative mass balances and components of mass balance for the spruce hillslope (a and c) and the beech hillslope (b and d) for sprinkling intensities of approximately 100 mm/h (a and b) and 60 mm/h. The soil water storage change (ΔSWS) scale is shifted so that initial SWS in each hillslope is 0. The vertical lines denote the end of sprinkling (duration = 1 h, except 60 mm/h under spruce, which lasted only 56 min).

in the soil region near the stagnic layer is always close to saturation, even during summer. Hence it takes only little water input to raise this soil layer to saturation and initiate macropore flow. Under spruce, the soil layer where the perched water table develops, is a layer of high fine root concentration (Schmid, 2002; Schmid and Kazda, 2002) and consequently of considerable water extraction. For the same stands, Schume et al. (2004) found that VWC in the top 30 cm of both stands is typically well below field capacity. Values in the growing seasons 2000 and 2001 ranged from 18% to 31% (Schume et al., 2004), with the spruce stand being on average 4–5% drier than the beech stand (differences become marginal under severe drought and amplified after rain). Because more water is needed to raise the water content in this layer to saturation and to initiate subsurface macropore flow, the runoff under spruce takes longer to initiate. A less active soil fauna (Scheu et al., 2003) and therefore fewer macropores might further enhance the time for runoff initiation under spruce.

Macropores transport water only over a limited distance in lateral direction (Anderson et al., 2008) until it finally has to re-enter the soil matrix. The soil matrix between sections of macropores needs to become saturated first, before water can flow into the next macropore. The effective flow velocity at a given point in time depends on the connectivity of the system (Noguchi et al., 1997; Weiler and McDonnell, 2007). Again, the different depths of quick

water infiltration and the differences in antecedent soil moisture conditions in and above the saturated zones between the two hillslopes influence, how they respond to intense rainfall. For the same reasons as mentioned above, the soil matrix between macropores in the spruce hillslope will take longer to reach saturation and to ultimately connect macropore flow laterally. On the contrary, the 50 and 70 cm soil layers under beech with lower porosity are always close to saturation and it will take only little additional water to connect macropore segments in these soil depths.

4.2. Runoff and interflow measurements

Several issues related to runoff collection in trenches and hillslope experiments in general suggest to interpret the observed hydrographs from both hillslopes with caution. The runoff collection from trenches impacts the natural flow conditions (Kirkby, 1978). This can involve the formation of saturated wedges at the margins of the trench or flow diverging from the trench (Freer et al., 2002). If the matric potential of the soil around the trench is more negative than at the trench surface, where the matric potential is at atmospheric pressure, then divergent flow conditions dominate. If the matric potential around the trench is higher than at the trench, e.g., when the soil matrix is saturated and water is under positive pressure, then water flow converges at the trench

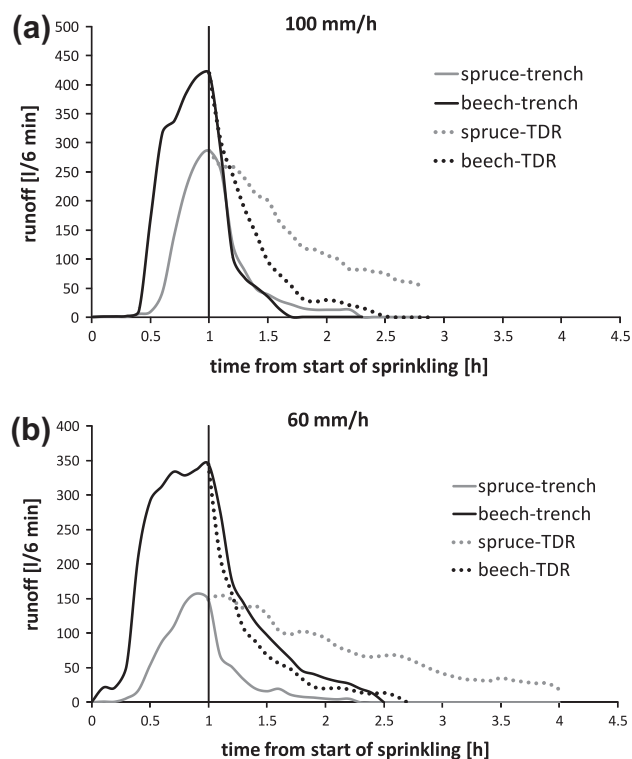


Fig. 8. Measured runoff (spruce-trench, beech-trench) and runoff with recession approximated from mean (48 TDR probes) decreases of soil water storage (spruce-TDR, beech-TDR) for sprinkling intensities of approximately 100 mm/h (a) and 60 mm/h (b) at the spruce and beech hillslopes. The vertical lines denote the end of sprinkling (duration = 1 h, note that 60 mm/h under spruce lasted only 56 min).

(Hillel, 1998; Jury and Horton, 2004). This offers an explanation why the mass balance in the 60 mm/h experiment under beech returns towards zero during the recession. These alterations of flow conditions suggest that when interpreting the tree species influence on hydrological processes, the runoff measurements at the trench wall should be given less weight than the observed soil water patterns.

The assumption of a no flux boundary condition at the sides of the hillslopes has been violated in all four experiments, which is nothing unusual in hillslope hydrology. Some water leaves the hillslope sideways through preferential pathways that are not slope parallel. Thereby the shallow network of large coarse roots of spruce could cause more water to move sideways than the deeper coarse root network of beech. Furthermore, the soil moisture patterns show that the perched water table which forms under spruce lies on top of dry soil, causing further water loss along the sides of the hillslope.

Under natural rainfall events of comparable intensity, which are not confined to a hillslope segment as our sprinkling experiments were, the VWCs at both hillslopes would be higher than what was observed in the sprinkling experiments. There are two reasons for that. First, it can be expected that the net flow through the sides of the hillslope segments would be balanced and not a net loss, thus enhancing the VWC. Second, there would be inflow from the top of the hillslope segment instead of a no flux boundary condition, which would likewise result in increased VWC. Since the mass balance was more or less closed in the 60 mm/h experiment at the beech hillslope, the VWC under real rainfall conditions under beech might only be slightly higher than observed. The higher VWC under natural conditions will shift hydrological thresholds, such as the amount of rain needed for saturation excess overland flow to occur (Kirkby, 1978). Saturation excess overland flow contributes substantial amounts of stormflow to the channels (Dunne and Black, 1970).

The risk of saturation excess overland flow is influenced by tree species. Under beech, the risk of saturation excess overland flow is low, because of the greater soil volume involved into the infiltration processes and therefore the higher storage capacity on top of the perched water table. Additionally the macropores in the top-soil of the beech stand (Fig. 9) turned out to be a more efficient drainage system than those in the spruce stand. For instance, the shallow interflow in the top 10 cm of the soil contributed 15.3% and 12.8% to the total measured runoffs in case of the beech stand, while these contributions were only 6.3% and 7.6% in the spruce stand for the two sprinkling intensities (Table 1).

Differences between measured and calculated SWS changes were more negative for the spruce hillslope segment (Fig. 7), which means that at the spruce hillslope more water was lost sideways than at the beech hillslope. Under natural rainfall conditions of high intensity with inflow from upslopes and balanced water fluxes perpendicular to the slope (i.e. sideways), the rain input threshold for saturation excess overland flow under spruce has to be set far lower than our experiments suggest. In fact, we observed signs of overland flow after heavy convective rainfall in the spruce stand several times during 2 years of field work in form of swept away litter. Overall, we conclude that the soil under spruce is more prone to saturation excess overland flow than the same soil under beech. From the sharp bend in the runoff recession limb, which is obvious in both experiments under beech, it can be concluded that there are two groups of large pores involved in the runoff process. One group that drains rapidly (e.g. mouse- and wormholes) and another group that drains more slowly. Under spruce, this sharp bend in the recession limb is hardly visible, which points towards a more gradual pore size distribution in the group of the large pores (i.e. the pores that drain between saturation and field capacity) with the very big pores missing.

Despite the evidence that differences in soil moisture dynamics and runoff processes between spruce and beech hillslopes are related to tree species, there are several limitations to our study that need to be considered. To minimize edge effects and given the limited amount of resources for our work, we opted to intensively monitor and sprinkle one large plot in each stand instead of extensively monitoring and sprinkling several small plots. Although the Kreisbach site was selected, because a pedon characterization based on 15 soil cores in each stand showed minimal variation compared to other potential sites, one large hillslope unlikely reflects the entire spatial variation of a stand. Hence, the differences that we observed at the hillslope scale might not be the same as the differences at the stand scale. In addition to tree root channels, soil properties such as thickness of mineral soil layers, texture class, or soil structure may play an important role for the development of preferential flow paths and thus may be responsible for part of the variation between the hillslopes.

The patterns of tree species effects on runoff processes in our study are likely specific to the soil type. Stagnic Cambisols have very high clay contents and drainage at high water contents almost exclusively happens in old root channels (Glatzel, 1982). As a result, the effect of different tree species may be easier to isolate in a Stagnic Cambisol than in other, not so heavily textured soils. Barros et al. (1999) showed that a system of macropores develops by internal soil mobilization and displacement and erosion until the soil structure is in equilibrium with its environment. Under natural rainfall, a tree canopy causes heterogeneous spatial patterns of throughfall that show some temporal persistence (Keim et al., 2005). For example, rainfall in a beech forest enters the soil preferentially near the stem whereas rainfall in a spruce forest reaches the soil preferentially between stems (Schume et al., 2004). Since the pore system in both stands is 'optimized' for natural conditions, our tree species effects inferred from a homogeneous application of water by sprinklers may

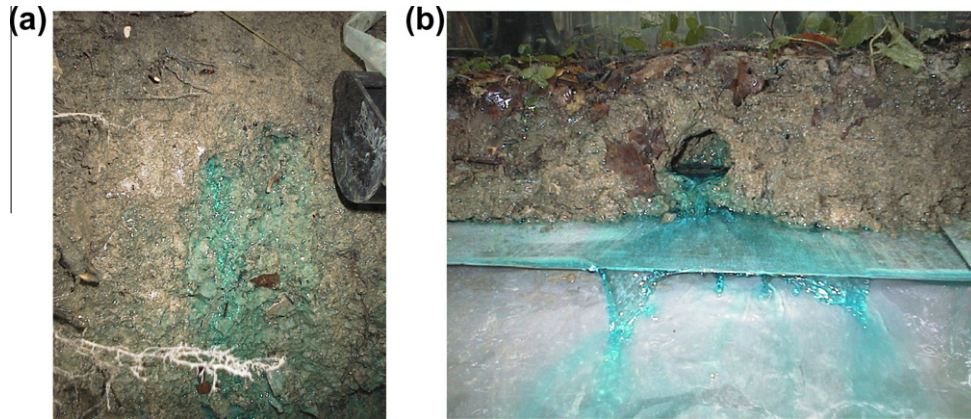


Fig. 9. Preferential flow in (a) earthworm channels and (b) a mousehole during the 100 mm/h sprinkling experiment at the beech hillslope. "Brilliant blue" was used as tracer.

differ from the effects of tree species on runoff processes under natural rainfall.

5. Conclusions

Our results suggest that water from intense rainfall under both beech and spruce infiltrates predominantly in macropores. Subsequently a perched water table rises towards the soil surface, under beech from the top of the dense, water logging soil layer (~60 cm deep) upwards, under spruce starting from the layer below the dense network of coarse roots (~30 cm deep). From the saturated zone in the perched water table lateral macropore flow gets initiated. Under beech the VWC directly above the stagnic soil layer is always close to saturation. Therefore it takes only little water input to bring the VWC to saturation, to initiate macropore flow and to connect segments of macropores. The beech hillslope exhibited a quicker runoff response to rainfall compared to a site stocked with spruce. Saturation excess overland flow is unlikely under beech because of the deeper lying perched water table, which considerably increases the available soil volume for water storage. Under natural rainfall conditions with inflow from upslopes, a site planted with spruce is prone to saturation excess overland flow, because of the shallower water table and the lower available soil volume for preferential flow.

These species dependent runoff processes are presumably caused by differences in the rooting systems of the two tree species and amplified by the soil fauna that is associated with beech and spruce. In deeper soil regions of beech forests, old root channels, i.e. potential macropores, are more frequent than in soils under spruce. Also, spruce stands are characterized by a less active, differently composed soil fauna consisting of smaller species groups and therefore fewer macro pores in the topsoil. All of which results in higher macro-porosity under beech and consequently in a faster runoff initiation, a steeper rising limb of the hydrograph, a steeper and sharply bent falling limb, but also a lower likelihood for saturation excess overland flow.

While we tried to isolate the effect of tree species from other controls on runoff generation by selecting sites with minimal variation in soil properties, we cannot eliminate that there are other factors that may have caused part of the variation between the spruce and the beech hillslope. To statistically separate the tree species influence from other factors would require a sample design with more than two large hillslopes. Patterns found in our study are likely specific to the heavily textured Stagnic Cambisols and the same tree species may alter runoff processes very differently on other soil types.

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References

- Alila, Y., Toews, D., 2003. Application of the distributed hydrology soil vegetation model to Redfish Creek, British Columbia: model evaluation using internal catchment data. *Hydrol. Process.* 17, 199–224.
- Alila, Y., Kuras, P.K., Schnorbus, M., Hudson, R., 2009. Forests and floods: a new paradigm sheds light on age-old controversies. *Water Resour. Res.* 45, W05205. doi:10.1029/2008wr007207.
- Anderson, A., Weiler, M., Alila, Y., Hudson, R., 2008. Dye staining and excavation of a lateral preferential flow network. *Hydrol. Earth Syst. Sci. Discuss.* 5, 1043–1065.
- Barros, A.P., Knapton, D., Wang, M., Kuo, C., 1999. Runoff in shallow soils under laboratory conditions. *J. Hydrol. Eng.* 4, 28.
- Berger, T., Hager, H., 2000. Physical top soil properties in pure stands of Norway spruce (*Picea abies*) and mixed species stands in Austria. *Forest Ecol. Manag.* 136, 159–172.
- Beven, K., Germann, P., 1982. Macropores and water flow in soils. *Water Resour. Res.* 18 (5), 1311–1325.
- Dekker, L.W., Ritsema, C.J., 2000. Wetting patterns and moisture variability in water repellent Dutch soils. *J. Hydrol.* 231–232, 148–164.
- Dunne, T., Black, R.D., 1970. An experimental investigation of runoff production in permeable soils. *Water Resour. Res.* 6 (2), 478–490.
- Ellenberg, H., 1996. *Vegetation Mitteleuropas mit den Alpen*. Verlag Eugen Ulmer, Stuttgart, Germany, 1096 pp.
- Freer, J. et al., 2002. The role of bedrock topography on subsurface storm flow. *Water Resour. Res.* 38 (12). doi:10.1029/2001WR000872.
- Glatzel, G., 1982. Standortkundliche Untersuchungen zur Frage der Bewirtschaftung von Stagnogleywaldstandorten im südöstlichen Österreich. *Centralblatt für das gesamte Forstwesen* 99 (2), 65–88.
- Hartge, K.H., Bläsing, D., Herklotz, K., 1983. Veränderungen des Bodengefüges unter dem Einfluss mehrjähriger Baumwurzeln. *Forstwissenschaftliches Centralblatt* 102, 99–110.
- Hibbert, A.R., Sopper, W.E., Lull, H.W., 1967. *Forest treatment effects in water yield*. Pergamon Press, The Pennsylvania State University, University Park, PA, August 29–September 10, 1965, pp. 527–543.
- Hillel, D., 1998. *Environmental Soil Physics*. Academic Press, San Diego, 771 pp.
- Hornbeck, J.W., Martin, C.W., Eagar, C., 1997. Summary of water yield experiments at Hubbard Brook Experimental Forest, New Hampshire. *Can. J. Forest Res.* 27, 2043–2052.
- Jost, G., Schume, H., Hager, H., 2004. Factors controlling soil water-recharge in a mixed European beech (*Fagus sylvatica* L.)–Norway spruce (*Picea abies* (L.) Karst.) stand. *Eur. J. Forest Res.* 93–104.
- Jost, G., Heuvelink, G.B.M., Papritz, A., 2005. Analysing the space–time distribution of soil water storage of a forest ecosystem using spatio-temporal kriging. *Geoderma* 128, 258–273.
- Jury, W.A., Horton, R., 2004. *Soil Physics*. John Wiley and Sons, 370 pp.
- Keim, R.F., Skaugset, A.E., Weiler, M., 2005. Temporal persistence of spatial patterns in throughfall. *J. Hydrol.* 314, 263–274.
- Kirkby, M.J., 1978. *Hillslope Hydrology*. Wiley, Chichester, 389 pp.

- Knight, D., Elliott, P., Anderson, J., Scholefield, D., 1992. The role of earthworms in managed, permanent pastures in Devon, England. *Soil Biol. Biochem.* 24, 1511–1517.
- Noguchi, S., Tsuboyama, Y., Sidle, R.C., Hosoda, I., 1997. Spatially distributed morphological characteristics of macropores in forest soils of Hitachi Ohta Experimental Watershed, Japan. *J. Forest Research* 2, 207–215.
- Scheu, S. et al., 2003. The soil fauna community in pure and mixed stands of beech and spruce of different age: trophic structure and structuring forces. *Oikos* 101, 225–238.
- Schmid, I., 2002. The influence of soil type and interspecific competition on the fine root system of Norway spruce and European beech. *Basic Appl. Ecol.* 3 (4), 339–346.
- Schmid, I., Kazda, M., 2001. Vertical distribution and radial growth of coarse roots in pure and mixed stands of *Fagus sylvatica* and *Picea abies*. *Can. J. Forest Res.* 31 (3), 539–548.
- Schmid, I., Kazda, M., 2002. Root distribution of Norway spruce in monospecific and mixed stands on different soils. *Forest Ecol. Manag.* 159 (1–2), 37–47.
- Schume, H., Jost, G., Hager, H., 2004. Soil water depletion and recharge patterns in mixed and pure forest stands of European beech and Norway spruce. *J. Hydrol.* 289, 258–274.
- Stednick, J.D., 1996. Monitoring the effects of timber harvest on annual water yield. *J. Hydrol.* 176, 79–95.
- Storck, P., Bowling, L., Wetherbee, P., Lettenmaier, D., 1998. Application of a GIS-based distributed hydrology model for prediction of forest harvest effects on peak stream flow in the Pacific Northwest. *Hydrol. Process.* 12, 889–904.
- Thyer, M., Beckers, J., Spittlehouse, D., Alila, Y., 2004. Diagnosing a distributed hydrologic model for two high-elevation forested catchments based on detailed stand- and basin-scale data. *Water Resour. Res.* 40. doi:10.1029/2003WR002414.
- Weiler, M., 2005. An infiltration model based on flow variability in macropores: development, sensitivity analysis and applications. *J. Hydrol.* 310 (1–4), 294–315.
- Weiler, M., McDonnell, J.J., 2007. Conceptualizing lateral preferential flow and flow networks and simulating the effects on gauged and ungauged hillslopes. *Water Resour. Res.* 43, 1–13.
- Weiler, M., Naef, F., 2003. Simulating surface and subsurface initiation of macropore flow. *J. Hydrol.* 273, 139–154.
- Wigmosta, M.S., Nijssen, B., Storck, P., Singh, V.P., Frevert, D.K., 2002. The distributed hydrology soil vegetation model. Water Resource Publications, Littleton, CO, 972 pp..
- Zinke, P.J., 1962. The pattern of influence of individual forest trees on soil properties. *Ecology* 43 (1), 130–133.